2023 SOAPS Invitational

Dynamic Planet



Science Olympiad at Penn State Invitational

Saturday, January 14th, 2023

The Pennsylvania State University, University Park, PA

Team Name:_____

Participant Names:_____

Total Score:____**/158** **Rank:**_____

Multiple Choice (1 point each)

1. The respective depth and width of a rectangular channel are 4 meters and 10 meters. Calculate the discharge of water through the channel if the bed slope is 1/600 and Chezy's constant is 50.
    a. 3.04 m^3/s
    b. 3.33 m^3/s
    c. 121.66 m^3/s
    d. 145.99 m^3/s
2. Which of the following effects does sediment have on aquatic organisms?
    a. Increasing small-order stream productivity
    b. Habitat destruction
    c. Increasing biodiversity
    d. Reducing competition among aggressive species
3. Streams with which characteristic are likely to have the most biodiversity?
    a. Strahler order of 0
    b. Lower Strahler order
    c. Higher Strahler order
    d. Strahler orders have no impact on biodiversity
4. Examples of point source pollution include all of these **except**:
    a. Fertilized farmland
    b. Smokestacks
    c. Discharge pipes
    d. Drainage ditches
5. Which of the following is **not** a common pollutant associated with point source pollution?
    a. Carbon dioxide
    b. Heavy metals
    c. Nitrogen oxides
    d. Ammonium nitrate
6. Between points A and B on a river, there is a river-parallel distance of 1.08 km and a straight-line distance of 0.60 km, what is the sinuosity of this river?
    a. 0.48
    b. 0.56
    c. 1.68
    d. 1.8
7. From **Question 6**, what is the corresponding channel type for this river?
    a. Braided
    b. Meandering
    c. Straight
    d. None of the above

8. Which type of rock would you expect to find a cavern in?
    a. Granite
    b. Shale
    c. Both
    d. None of the above
9. On the outside of a meander curve, when compared to the inside, the river's velocity is _____.
    a. Equal
    b. Lower
    c. Higher
    d. Variable
10. What identifies the "competence" of a stream?
    a. Stream velocity
    b. The maximum particle size the stream carries
    c. The volume of water in a particular section of a stream
    d. The quantity of sediment carried by a stream
11. Coarse sediment settles _____ and fine sediment settles _____.
    a. upstream, downstream
    b. downstream, upstream
    c. on the shore, in the streambed
    d. in the streambed, on the shore
12. What is the name of the device used to measure evaporation?
    a. Turbidimeter
    b. Hygrometer
    c. Atmometer
    d. Lysimeter
13. Which of the following factors affect water infiltration of a formation?
    a. Wind velocity
    b. Type of soil medium
    c. Soil acidity
    d. Vegetative cover
    e. Soil compaction
    f. A, B, D, E
    g. A, C
    h. B, D, E
    i. B, C, E
    j. D, E
    k. All of the above
    l. None of the above

14. How are oxbow lakes formed?
    a. Tectonic activity
    b. Erosion and deposition
    c. High stream velocity
    d. Flooding
15. Where could one find a knickpoint?
    a. Niagara Falls on the Canada-United States border
    b. Bridalveil Fall in Yosemite
    c. Dry Falls in Washington
    d. All of the above
    e. None of the above
16. What direction do knickpoints migrate, why?
    a. Upstream due to erosion
    b. Upstream due to sediment deposition
    c. Downstream due to erosion
    d. Downstream due to sediment deposition
17. What causes the formation of knickpoints?
    a. Isostatic rebound
    b. Tectonic uplifting
    c. Decreased ocean water influx
    d. All of the above
18. Which of the following sediments forms the most effective aquiclude?
    a. Sand
    b. Clay
    c. Gravel
    d. Sandstone
19. What water feature do dotted lines indicate on a USGS topographic map?
    a. Intermittent streams
    b. Perennial streams
    c. Disappearing streams
    d. Perennial rivers
20. What combination of temperature, density, and oxygen content best describes water found in a stratified lake's hypolimnion (relative to other layers)?
    a. Low temperature, high density, high oxygen content
    b. High temperature, high density, low oxygen content
    c. Low temperature, low density, high oxygen content
    d. Low temperature, high density, low oxygen content

21. Which of the following causes river rejuvenation?
    a) Rise in sea level
    b) Uplift
    c) Heavy Rain
    d) Bifurcation
22. Which of the following is **not** a form of accordant drainage?
    a) Consequent
    b) Antecedent
    c) Resequent
    d) Subsequent
    e) Insequent
23. Which of the following correctly orders the listed US watersheds by area from largest to smallest?
    a) Mid Atlantic Region, Great Lakes Region, Ohio Region, New England Region
    b) Ohio Region, Mid Atlantic Region, Great Lakes Region, New England Region
    c) Mid Atlantic Region, Ohio Region, New England Region, Great Lakes Region
    d) Ohio Region, Great Lakes Region, Mid Atlantic Region, New England Region
24. Which of the following is **not** a zone of a stream?
    a) Fluvial zone
    b) Deposition zone
    c) Headwater zone
    d) Transfer zone
25. Which of the following is **not** a gauge used to measure different characteristics of a stream or river?
    a. Vertical pipe gauge
    b. Staff gauge
    c. Henderson gauge
    d. Wire weight gauge
26. Which of the following correctly orders the steps of watershed eutrophication?
    a. Algal blooms form, excess nutrients wash into watershed, algae blocks sunlight from reaching the bottom of the affected body, algae uses up dissolved oxygen, water becomes dead zone
    b. Excess nutrients wash into watershed, algal blooms form, algae blocks sunlight from reaching the bottom of the affected body, algae uses up dissolved oxygen, water becomes dead zone
    c. Excess nutrients wash into watershed, algal blooms form, algae uses up dissolved oxygen, algae blocks sunlight from reaching the bottom of the affected body, water becomes dead zone
    d. Algal blooms form, excess nutrients wash into watershed, algae uses up dissolved oxygen, algae blocks sunlight from reaching the bottom of the affected body, water becomes dead zone

27. Which definition correctly identifies the differences between dimictic, oligomictic, and polymictic lakes?
    a. Dimictic lakes have turnover in the winter and stratify through the warmer seasons, Oligomictic lakes have turnover in the summer and stratify in the colder seasons, Polymictic lakes have turnover in the fall and spring with stratification in winter and summer
    b. Dimictic lakes have turnover in the fall and spring with stratification in winter and summer, Oligomictic lakes have turnover in the winter and stratify through the warmer seasons, Polymictic lakes have turnover in the summer and stratify in the colder seasons
    c. Dimictic lakes have turnover in the fall and spring with stratification in winter and summer, Oligomictic lakes have irregular turnover and remain generally warm, Polymictic lakes have frequent turnover with little distinction in season
    d. Dimictic lakes have irregular turnover and remain generally warm, Oligomictic lakes have turnover in the fall and spring with stratification in winter and summer, Polymictic lakes have frequent turnover with little distinction in season
28. Which of the following lakes did **not** form through volcanic activity?
    a. Lonar Lake
    b. Crater Lake
    c. Lake Toba
    d. Lake Nyos
29. Which of the following contribute to the formation of sinkholes? Select all that apply:
    a. Excess withdrawal of groundwater
    b. Cave roof collapses
    c. Limestone bedrock erosion
    d. Water table lowers
30. Which of the following materials has the lowest porosity?
    a. Sand
    b. Clay
    c. Silt
    d. Gravel
31. What is the maximum salinity in freshwater?
    a. 10 ppm
    b. 100 ppm
    c. 500 ppm
    d. 1000 ppm
32. Which of these organisms **do not** live in a freshwater environment?
    a. Mayfly
    b. Asian carp
    c. Zooxanthellae
    d. Water lettuce

33. What factor has the greatest impact on drinking water quality?
    a. Total dissolved solids
    b. Temperature
    c. pH
    d. Dissolved oxygen
34. Which of the following acids dissolves limestone?
    a) Hydrochloric acid
    b) Nitric acid
    c) Sulfuric acid
    d) Carbonic acid
35. Which of the following is **not** a type of bog?
    a) Blanket bog
    b) Quaking bog
    c) Valley bog
    d) Field bog
36. Which of the following wetlands is known for its lack of nutrients and inability to support large plant life?
    a) Fens
    b) Swamps
    c) Marshes
    d) Bogs
37. What type of sediment load is deposited first when a river enters the ocean?
    a. Dissolved load
    b. Bed load
    c. Suspended load
    d. None of the above
38. What desalination method uses high pressure to force saltwater through a membrane filter?
    a. Diffusion
    b. Distillation
    c. Active Transport
    d. Reverse osmosis
39. What causes most coastal pollution in the United States?
    a. Overfishing
    b. Recreational activity
    c. Industrial waste and sewage
    d. Agricultural runoff
40. Terrain that does not feature sinkholes and disappearing streams is called Karst topography.
    a. True
    b. False

Identification

41. Identify the following waterfalls with their appropriate names: (1 point each):

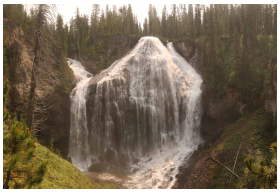
a. 

b. 

c. 

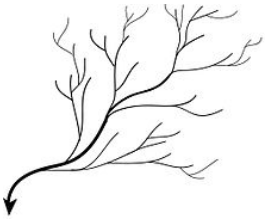
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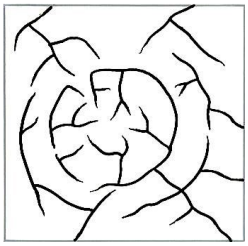
e. 

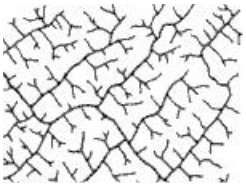
f. 

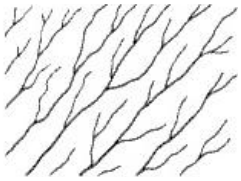
g. 

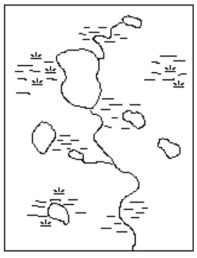
42. Identify the drainage pattern along with its associated topography (2 points each):

a. 

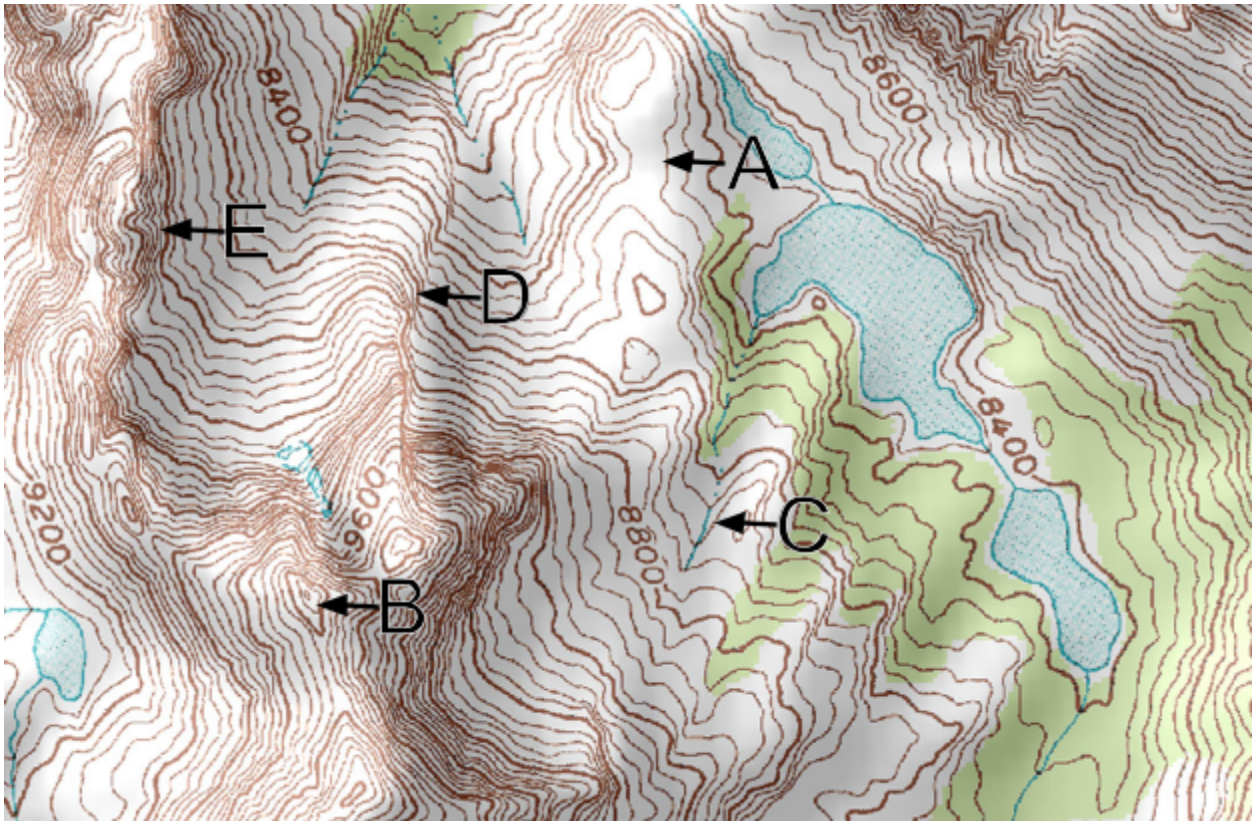
b. 

c. 

d. 

e. 

43. Identify basic topographic landforms according to the labeled map below (1 point each).



a. _____

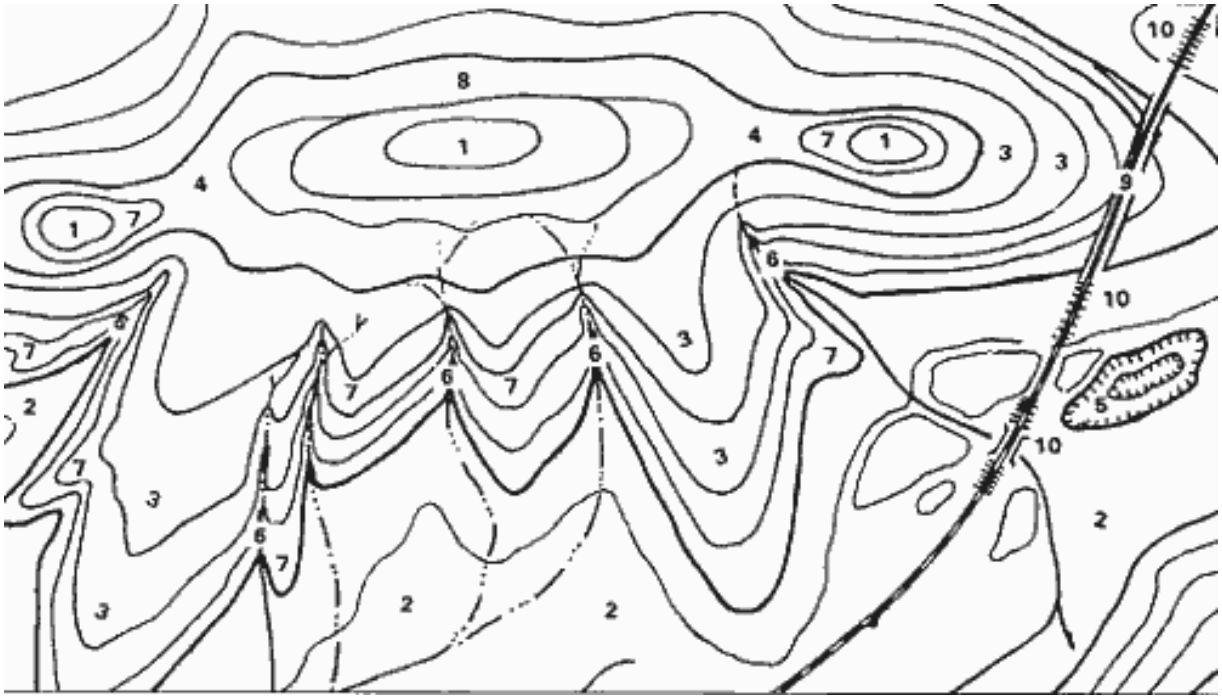
b. _____

c. _____

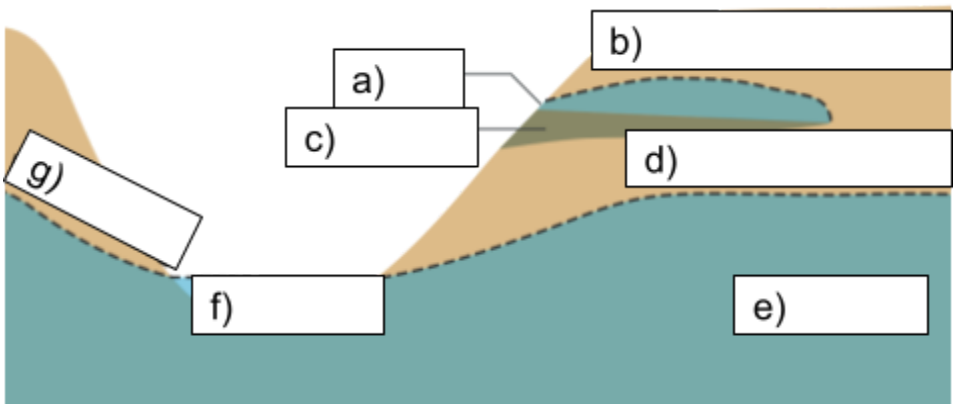
d. _____

e. The contour lines here indicate a _____ slope.

44. Identify additional basic topographic landforms according to the map below (1 point each).



45. Identify the basic elements of the surface-water diagram (1 point each).



Open-Ended

46. What differentiates aquifers, aquitards, aquicludes, and aquifuges? Describe the sediment forms that characterize each of these geological formations (8 points).

47. Order the following materials from lowest to highest porosity: salts, gravel, silt, clay, and sand (5 points).

48. What are ways that humans impact groundwater quality (6 points total)?

    a. Identify at least three human actions that impact groundwater quality (3 points).

    b. Explain the effects of these actions on groundwater quality (3 points).

49. Sketch the water/hydrologic cycle, labeling the following key terms: water storage in ice and snow, precipitation, snowmelt runoff to streams, infiltration, streamflow, evapotranspiration, condensation, evaporation, water storage in oceans, freshwater storage, spring, groundwater discharge, groundwater storage, sublimation, water storage in the atmosphere (15 points).

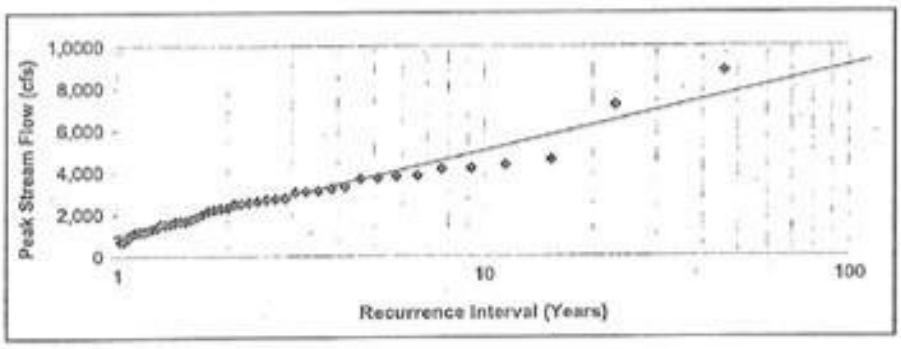
50. The data table below contains the data set from the Macatawa River ranked according to peak stream flow with AEP and T calculated. Use data analysis skills and knowledge of hydrology to answer the questions found on the next page (45 points total).

Table 8.1. Peak Flows for the Macatawa River

Rank	Year	Recurrence Interval (T)	Peak flow (Q) (cfs)	AEP
1	1997	46	8810	0.02173913
2	1981	23	7220	0.043478261
3	1982	15.33333333	4600	0.065217391
4	1996	11.5	4340	0.086956522
5	1979	9.2	4180	0.108695652
6	1989	7.666666667	4150	0.130434783
7	1978	6.571428571	3830	0.152173913
8	1994	5.75	3800	0.173913043
9	1973	5.111111111	3710	0.195652174
10	1985	4.6	3690	0.217391304
11	1980	4.181818182	3310	0.239130435
12	1983	3.833333333	3200	0.260869565
13	1984	3.538461538	3110	0.282608696
14	1991	3.285714286	3080	0.304347826
15	1986	3.066666667	3050	0.326086957
16	1966	2.875	2760	0.347826087
17	1976	2.705882353	2730	0.369565217
18	2000	2.555555556	2710	0.391304348
19	2001	2.421052632	2650	0.413043478
20	1974	2.3	2570	0.434782609
21	1965	2.19047619	2520	0.456521739
22	1993	2.090909091	2500	0.47826087
23	2002	2	2330	0.5
24	1999	1.916666667	2270	0.52173913
25	1995	1.84	2210	0.543478261
26	1971	1.769230769	2170	0.565217391
27	1972	1.703703704	2010	0.586956522
28	2005	1.642857143	1870	0.608695652
29	1963	1.586206897	1780	0.630434783
30	1962	1.533333333	1690	0.652173913
31	1967	1.483870968	1690	0.673913043
32	1975	1.4375	1660	0.695652174
33	1970	1.393939394	1570	0.717391304
34	2004	1.352941176	1530	0.739130435
35	1992	1.314285714	1520	0.760869565
36	1968	1.277777778	1370	0.782608696
37	1988	1.243243243	1320	0.804347826
38	1987	1.210526316	1240	0.826086957
39	1990	1.179487179	1220	0.847826087
40	1998	1.15	1180	0.869565217
41	1964	1.121951122	1130	0.891304348
42	2003	1.095238095	1040	0.913043478
43	1969	1.069767442	850	0.934782609
44	1977	1.045454545	728	0.956521739
45	1961	1.022222222	719	0.97826087

Note: AEP = annual exceedance probability; cfs = cubic feet per second.

Figure 8.1. Macatawa River Flood Frequency Curve



a) Using the flood frequency graph for the river, estimate the discharges (peak streamflow) of 10-year, 25-year, 50-year, and 100-year floods (2 points each)
    I. 10-year
    II. 25-year
    III. 50-year
    IV. 100-year
b) What is the recurrence interval of a 50-year flood (5 points)?
c) Approximately 1,500 CFS is needed to move sediment. Find the recurrence interval and the probability of this flow in any given year (10 points).
d) What are the effects of increased sediment loads being transported down the river (10 points)?
e) Since the USGS has been collecting data for the watershed, what was the highest peak streamflow and in what year did it occur? What are the probability and recurrence interval of this discharge (12 points)?